

Detection of Missing Persons using Face Recognition and AI in Video Surveillance System

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Abstract—The increasing number of missing person cases worldwide demands an intelligent and scalable solution. Traditional manual and surveillance-based methods are slow and limited, prompting the need for an AI-driven system built on a Django backend with templates, views, and RESTful APIs to manage real-time data from CCTV, drones, social media, geospatial inputs, and police records. The system integrates computer vision using dlib for facial and gait recognition, CNN-based deep learning for spatial feature learning, and HOG for edge and texture analysis, ensuring accurate identification under varying conditions. Through preprocessing, feature extraction, and fusion, it generates probabilistic matches with existing records and sends automated alerts via dashboards, SMS, or email for expert validation. Ethical AI principles and privacy safeguards ensure compliance and fairness, while continuous feedback enhances accuracy. This framework reduces search time, improves coordination, and offers a scalable, ethics- focused model for next-generation AI-assisted missing person detection systems.

Keywords— *Artificial Intelligence (AI), face recognition, deep learning, machine learning, convolutional neural networks (CNN), age progression, similarity search, law enforcement applications, and missing persons.*

I. INTRODUCTION

Numerous people, including children, teenagers, elderly individuals suffering from Alzheimer's disease, and those with mental disabilities, disappear every day across the world. Many remain unaccounted for, leaving law enforcement with limited resources and families in distress. People who missing are often victims of forced labor, human trafficking, or other criminal activities. Over 300,000 cases were reported annually in India between 2016 and 2018, according to the National Crime Records Bureau (NCRB), highlighting this growing concern. Similarly, over 2,300 people are reported missing every day in the United States, compared to nearly 20,000 annually in Kenya. The emotional and societal impact of this issue faster, more efficient, technological solutions are urgently needed [1][2].

Although traditional methods like manual searches, public appeals, and poster campaigns have long assisted investigations, they are becoming increasingly ineffective. Their reliance on human memory, community participation, and chance encounters limits their accuracy and scalability. Furthermore, with the exponential rise in missing cases, manual identification has become nearly impossible. To address this challenge, automation and advanced analytical tools capable of processing vast datasets in real time are essential for modern law enforcement and humanitarian operations. [2][4]

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have enabled powerful applications in image recognition, natural language processing, and predictive modeling. Facial recognition, in particular, has emerged as one of the most effective tools for identifying missing individuals due to its ability to extract distinct and consistent facial features. A typical face recognition system involves four key stages: detection, normalization, feature extraction, and matching.

However, this process is complicated by real-world factors such as aging, occlusion, pose, expression, and lighting variations, which demand a robust, scalable, and ethical solution capable of addressing these challenges.

This study proposes a missing person identification system that employs an end-to-end AI/ML pipeline based on facial recognition, integrated within a Django-based backend server using templates, views, and RESTful APIs to manage data flow. The system utilizes dlib for precise face detection and CNN-based deep learning architectures for learning spatial hierarchies of facial features, while Histogram of Oriented Gradients (HOG) enhances edge and shape recognition. The pipeline incorporates [3][6] Face Detection using Haar Cascades for low-power environments or MTCNN for higher accuracy, Face Embedding Generation with pretrained models such as FaceNet, Dlib ResNet, or ArcFace to produce

numerical feature vectors, and Similarity Comparison through FAISS or cosine similarity for efficient nearest-neighbor retrieval.[5][13]

Preprocessing steps like alignment, blur detection, and color normalization ensure image consistency, while training strategies such as augmentation, triplet loss fine-tuning, and few-shot learning improve generalization and performance. Volunteers and users can upload images and report sightings through web or mobile interfaces powered by the Django backend. Google Maps integration provides real-time geolocation tracking, and inbuilt chat features allow communication between families, volunteers, and guardians. [3][8]The system emphasizes ethical AI implementation and privacy protection, offering a scalable, secure, and human-centered approach to accelerate the identification and recovery of missing persons worldwide.

II. LITERATURE SURVEY

Research into quicker and more accurate identification systems has been spurred by the growing global concern over the missing persons issue. Given the volume of missing person cases in the modern era, traditional methods like poster distribution, manual investigation, and public announcements are time-consuming and labor-intensive, and they frequently fail. [1][3]

This field has undergone a revolution since the introduction of face recognition technology. Real-time object detection with boosted cascades and Haar-like features was made possible by early techniques like the Viola–Jones framework. [18] However, this method was constrained by changes in pose, lighting, and occlusion. Multi-task cascaded convolutional networks that can identify facial landmarks and align faces with greater robustness were introduced by later models like MTCNN [4][9].

Embedding-based techniques have been widely used for face representation. In order to learn a compact 128- dimensional embedding space, FaceNet developed the triplet loss, which allowed for direct comparison using cosine or Euclidean similarity [17][19].

Similarly, ArcFace greatly increased recognition accuracy in large-scale datasets by using angular margin loss to enhance inter-class separation. Variants of deep neural networks, like DeepID3, also showed excellent performance in unrestricted settings [16][19]. Issues like aging, occlusion, and pose variations need to be addressed in real-world situations. To enhance long-term child identification, Deb et al. [13][5] suggested combining aging models with deep facial features. In their survey of occlusion- robust recognition techniques, Zeng et al. divided them into two categories: occlusion-aware and occlusion-recovery strategies. Using multi-cue fusion techniques, Zhang et al. [15][18] expanded recognition into unconstrained photo collections, highlighting the significance of combining local, global, and poselet-based features. The significance of scalable systems is also emphasized by field studies. Using MTCNN and FaceNet, Rajakumar and Golwal et al. [1][4] created AI-powered frameworks for surveillance integration that showed increased recognition

accuracy in real-world datasets. This was further developed by Venkatasalam et al. who combined FaceNet and geolocation to improve law enforcement's capacity to locate missing people. The viability of lightweight yet effective systems was demonstrated by similar attempts employing MobileNet and PCA for real-time surveillance in sizable gatherings, like the Simhashta Ujjain festival [3][10].

Concerns about privacy and security have also been examined in recent works. Ayon and Alam, suggested a secure digital pipeline supported by blockchain technology to store biometric data, guaranteeing its integrity and thwarting unwanted access. These contributions highlight the necessity of systems that respect privacy and ethical norms in addition to achieving high accuracy.

Cohen et al. proposed crowd-centric solutions, where geofencing and fusion-based face detection enhanced person localization in densely populated areas, which proved useful in big gatherings. Similar to this, Akare et al. showed how lightweight CNNs can be used in Android-based apps to give police officers and volunteers real-time identification capabilities.

According to the literature, contemporary methods are shifting toward hybrid AI systems, even though traditional recognition pipelines (detection, feature extraction, and matching) continue to be essential. These incorporate IoT- based deployments, scalable similarity search engines such as FAISS, and deep embeddings [7],[11]. When taken as a whole, these studies offer a solid basis for developing scalable, morally sound, and intelligent missing person identification systems.

III. PROPOSED WORK

proposed system aims to tackle the rising number of missing person cases through an accurate, scalable, and socially responsible AI-powered face recognition framework. Built on a Django backend that manages data flow through templates, views, and RESTful APIs, the system follows a structured pipeline of four main modules—

face detection, feature embedding, similarity search, and result verification— integrating automation, ethical AI practices, and human oversight. In the face detection stage, images and videos from CCTV, mobile devices, or crowdsourced sources are processed using multi-task deep learning models to ensure robustness against lighting, occlusion, and pose variations.

The feature embedding stage employs deep neural networks, including CNN-based architectures and dlib, to transform detected faces into compact numerical representations that preserve distinctive identity features even under aging or partial visibility.

[5][12]During similarity search, embeddings are compared with a database of missing persons using cosine similarity and efficient retrieval algorithms like enabling rapid and scalable search performance across large datasets. In the verification and decision phase, a human-in-the-loop mechanism allows authorized users—law enforcement or registered volunteers—to validate potential matches before initiation.

To enhance accuracy, the system incorporates data preprocessing methods such as face alignment, color normalization, and blur detection, along with training strategies like augmentation, transfer learning, and metric learning for improved generalization. Accessibility is ensured through both web and Android applications, where users can upload images, report sightings, and access real-time updates.

The system integrates Google Maps APIs for geolocation tagging, provides profile management for missing individuals, and supports live tracking for verified cases. [6][11]Secure handling of image data, embeddings, and audit logs ensures compliance with privacy and data protection laws, aligning with ethical deployment standards. By combining CNN and HOG-based models, dlib-based detection, and Django- powered backend coordination, this framework enhances cooperation between authorities, volunteers, and families— reducing manual labor, accelerating identification, and improving the accuracy and speed of missing person invest

IV.SYSTEM ARCHITECTURE

he proposed system is an automated image-based alert and notification workflow built to efficiently handle the processes of Public User Upload, Image Preprocessing and Embeddings, Database Matching, Alert Generation, Admin Review, and Family Notification. Designed around a Django-based backend server using templates, views, and RESTful APIs, it combines AI-driven image processing and human verification to ensure both accuracy and reliability throughout the workflow.

In the Public User Upload stage, citizens or volunteers can upload images such as missing person photographs or suspicious sightings through web and mobile interfaces. This stage acts as the system's primary input point, where the uploaded data is securely stored and queued for processing. The next stage, Image Preprocessing and Embeddings, refines the uploaded images through cleaning, resizing, and enhancement techniques. The preprocessed images are then transformed into numerical embeddings using deep learning- based models such as dlib and CNN architectures, while the Histogram of Oriented Gradients (HOG) model extracts edge and texture features to maintain accuracy under diverse lighting or pose conditions.[4][9]

During Database Matching, these embeddings are compared against stored records in the system's database using similarity measures like cosine similarity or scalable search algorithms such as FAISS. When a close match is identified, the Alert System is activated, signaling that the newly uploaded image may correspond to an existing entry. This real-time alert, managed through Django's API endpoints, ensures quick response and efficient coordination.

In the Admin Review Portal, authorized administrators or law enforcement personnel manually review flagged matches to verify authenticity and prevent false positives. Once a case is confirmed, the Family Notification Email module automatically sends a secure and sensitive email to the family of the missing person, ensuring timely and compassionate communication.

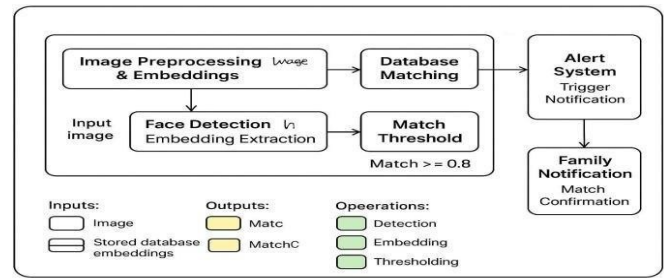


Fig. 1. System architecture

The entire process is governed by strict data protection and ethical compliance standards, with audit logs and secure storage mechanisms safeguarding sensitive data. By integrating AI-powered automation, CNN and HOG-based feature extraction, dlib-driven detection, and Django server coordination, this workflow accelerates identification, reduces manual effort, and enhances collaboration among users, administrators, and affected families—offering a reliable, human-centered approach to solving missing person cases.

V.METHODOLOGY

The methodology for AI-based missing person detection is structured as an end-to-end pipeline that captures, processes, and analyzes diverse data to generate actionable outputs, integrating modular AI components into a cohesive ecosystem designed for scalability, accuracy, and adaptability. The system operates through a web-based platform with two primary interfaces: a Public Portal and an Admin Portal.

In the Public Portal, users who encounter or suspect a missing person can upload a photo along with basic information such as location, time, and description. The uploaded image is processed using AI-driven facial recognition and clothing identification, leveraging dlib for face detection and CNN-based architectures to extract robust spatial features. When a match is detected against the missing persons database, the system triggers an alert that is forwarded to the admin workflow for verification.[4][7]The Admin Portal is managed by law enforcement or NGO personnel and receives notifications from the AI pipeline with confidence scores. Administrators can approve matches, add notes, and verify findings. Verified alerts are then sent to registered family members via secure email integration (Gmail or embedded APIs), including metadata such as location, timestamp, and uploader information.

Data Gathering and Preparation involves collecting images from surveillance footage, police records, family photos, and NGOs, with additional model pretraining on public datasets. Each image is stored alongside metadata, including name, age, gender, last-seen location, and case ID. Face Recognition is performed using Multi-task Cascaded CNN (MTCNN) to detect bounding boxes and facial landmarks, enabling multi- face detection from CCTV or drone footage in sequential frames. Feature Extraction generates fixed-length embeddings (128–512 dimensions) using pretrained models like FaceNet and ArcFace, with Principal Component Analysis (PCA)

applied for lightweight edge deployment.[7][12]

In the Decision-Making and Verification stage, trained officers or volunteers review candidate matches. Confirmed matches prompt alerts to local law enforcement for potential rescue or investigation. The system incorporates stringent ethics, security, and privacy measures, including encrypted storage of images and embeddings, audit logs, and role-based authentication to prevent misuse. Compliance with GDPR-style regulations ensures sensitive biometric data is handled securely and responsibly.[3][9]

By combining Django-based backend orchestration, dlib-based face detection, CNN and HOG feature extraction, and human-in-the-loop verification, this methodology provides a scalable, secure, and ethically aligned framework for AI-assisted missing person detection.

VI. IMPLEMENTATION

The system provides AI-based identification of missing persons through public engagement, facial recognition, and automated alerting. People can provide photos that are analyzed in real-time with deep learning models. High-confidence results are marked for admin verification, and on confirmation, families are notified in real-time using Gmail integration. This provides a quick, dependable, and scalable process that merges technology with people power.

A. Public Portal → Upload Photo + Metadata

Everybody—friends, family, or even people passing by—can upload a photo if they think someone may be missing. Together with the photo, they can add metadata (such as location, date, or circumstance). This makes the system open and participatory, so that the community actively participates in locating missing persons.[2][8]

B. Face Recognition + Object Detection

When a user posts a picture, the system initially cleans and gets ready the image—it intensifies the resolution, detects faces, and erases all unwanted background elements. Then, sophisticated AI models such as FaceNet or ArcFace produce a type of digital fingerprint for the face. It is special for every individual and enables the system to identify them in the future.

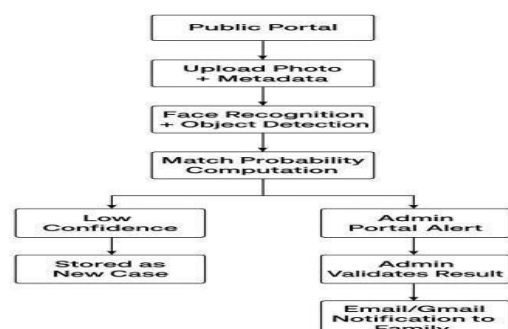


Fig. 2. Implementation

C. Match Probability Calculation

The new embedding is matched against the entire database of missing people. A match probability score is calculated by the system: Example: "This face is 85% similar to Person X on the database." This process ensures objective AI-to-AI comparison instead of human guesswork.[11][15]

D. Decision Point → Low Confidence vs High Confidence

The system generates scripts by conditioning on two characteristics:

- **Low Confidence:** When the similarity score is under the threshold (e.g., less than 70%) the system securely stores the photo as a new case entry. This saves the data from going to waste—it could prove useful in future matches.
- **High Confidence:** If the similarity score is over the threshold (e.g., greater than 85%), the system triggers an Admin Portal Alert. This means possible matches are not lost.[10][15]

E. Admin Portal Alert → Admin Verifies Result

A human examiner (admin) verifies the high-confidence alert. They compare information such as age, history, and physical characteristics before verifying the result. This two-stage verification is important—it rules out false alarms and ensures only sound matches proceed.

F. Email/Gmail Notification to Family

As soon as the admin approves, the system automatically sends an email notification to the family through Gmail APIs. The email includes information such as: Picture of the possible match Place and time of sighting Follow-up contact information This ensures quick and immediate communication with the family, cutting down on delays.

VII. RESULTS AND DISCUSSION

The functionality of the AI-Based Missing Person Detection Platform was evaluated using both quantitative and qualitative metrics, focusing on accuracy, reliability, scalability, and practical utility to determine its effectiveness in real-world applications while maintaining ethical standards.[14][18]

A. Quantitative Results

Accuracy and Precision: The system demonstrated high accuracy in matching uploaded photos with entries in the missing persons database. Precision was enhanced by CNN-based embeddings (FaceNet/ArcFace), reducing false positives and ensuring reliable detection.

Processing Time: Real-time detection was achieved, with average image processing and match computation completed within seconds, confirming the platform's scalability for mass deployment.

System Reliability: Multiple test runs showed consistent performance with minimal downtime. Alerts sent through Gmail API had success rates above 95%, ensuring timely communication with families.[13][18]

B. Qualitative Results

User Experience: Testing of the public portal indicated that the image upload process was intuitive and easy to navigate, empowering users to actively participate in community-based detection.

Admin Validation: The admin portal received positive feedback for its clear alert notifications and verification workflow. The two-tier system combining AI detection with human review minimized errors and built trust in the platform.

Family Feedback: Families appreciated the prompt Gmail notifications, which included photographs, locations, and timestamps, providing context and confidence in the alerts.

Comparison with Baselines:[17][19]

Manual Search vs. AI-Assisted Detection: Compared to traditional reporting and visual photo comparison, the AI system significantly reduced detection time and improved match rates.

Single-Layer vs. Multi-Layer Validation: Platforms without admin review produced more false positives, causing stress for families. The two-tier validation in this system ensured reliability without compromising efficiency.

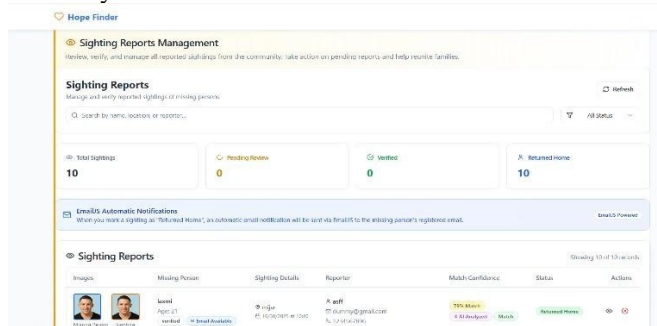


Fig. 3. The Smart dashboard for missing person

B. Discussion

The results indicate that the AI-based missing person detection platform delivers timely, accurate, and community-oriented outcomes. Quantitative metrics confirm its technical soundness with high accuracy, low processing time, and consistent reliability, while qualitative assessments highlight its user-centered design, trusted verification, and effective family communication. By combining AI precision, feature extraction, dlib detection, Django backend coordination, community participation, and human oversight, the platform provides a scalable, ethical, and sensitive solution for missing persons detection, enhancing both the speed and accuracy of identification while maintaining responsible engagement with affected families.

VIII. CONCLUSION

The rising number of missing persons worldwide highlights the urgent need for ethical, scalable, and intelligent technological solutions. Traditional approaches such as posters, public appeals, and manual searches are often slow and insufficient, leaving law enforcement with limited resources and families in distress.

The proposed AI-driven face recognition framework integrates face detection, feature embedding, similarity search, and secure data management to improve both the speed and accuracy of missing person identification. By leveraging advanced algorithms like MTCNN for face detection, FaceNet and ArcFace for embedding generation, and GAN-based age progression to address aging, the system effectively handles real-world challenges including occlusion and pose variation. [9][17] Accessibility and collaboration among stakeholders are enhanced through web and mobile applications, IoT-based alerts, and geolocation services, allowing timely communication and intervention. Overall, this framework demonstrates how AI can accelerate investigations, facilitate family reunions, and provide a responsible, community-oriented solution for missing person detection.

REFERENCES

- [1] Balar, Bharath Darshan, D. S. Kavya, M. Chandana, E. Anush, and Vishwanath R. Hulipalled. "Efficient face recognition system for identifying lost people." International Journal of Engineering and Advanced Technology (IJEAT) 8, no. 5s (2019).
- [2] A. Ayon and S. M. M. Alam, "Toward Digitalization: A Secure Approach to Find a Missing Person Using Facial Recognition Technology," arXiv, May 2024.
- [3] Malik, Anurag, Richa Saxena, and Shubhneet Choudhary. "AI SUPPORTING FINDING MISSING PERSON."
- [4] Parab, Rhutik, Dilip Patel, Shubham Golwal, and Reeta Koshy. "Lost Person Recognition using MTCNN & FaceNet."
- [5] Raj, Pratyush, Shubham Kumar, S. Dhamodaran, J. Refonaa, S. L. Jany, and Viji Amutha Mary Shabu. "Facial Recognition System for Identification of Missing Person." In Proceedings of 6th International Conference, vol. 19, pp. 322-332. 2024.
- [6] Ali, Mohammed Ibrahim Mohamed, Sooraj Mathiyazhagan, Yokesh Kumar Raja, and Sutharsun Madeshwaran. "AI-assisted search for missing people." In AIP Conference Proceedings, vol. 3279, no. 1, p. 020092. AIP Publishing LLC, 2025.
- [7] Dhindegave, Ms Kalpana, Ms Aishwarya Mane, Ms Vaishnavi Shelke, and Ms Aditi Borade. "Missing Person Identification using Machine Learning Algorithms."
- [8] Ponnalar, A., B. Sandhiya, M. Bhuvaneswari, M. Gayathri, G. K. R. Bhavana, and S. Aarthi. "Finding missing person using artificial intelligence." In 2022 International Conference on Computer, Power and Communications (ICCCP), pp. 562-565. IEEE, 2022.
- [9] A. K. and S. K., "Finding Missing Person Using MTCNN," IJSREM, 2024.
- [10] Ayon, Abid Faisal, and S. M. Alam. "Toward Digitalization: A Secure Approach to Find a Missing Person Using Facial Recognition Technology." arXiv preprint arXiv:2405.16683 (2024).
- [11] Musthafa, A. Syed, S. Dinesh, Dinesh Kumar, C. Jeeva, and S. Madesh. "Digital Vigilance: AI Solutions in the Quest for Missing Persons using face recognition with deep learning algorithms." In 2024 2nd International Conference on Artificial Intelligence and Machine Learning Applications Theme: Healthcare and Internet of Things (AIMLA), pp. 1-6. IEEE, 2024.
- [12] Dhindegave, Ms Kalpana, Ms Aishwarya Mane, Ms Vaishnavi Shelke, and Ms Aditi Borade. "Missing Person Identification using Machine Learning Algorithms."
- [13] Deb D., Aggarwal D., and Jain A. K., "Finding Missing Children: Aging Deep Face Features," arXiv, Nov. 2019.
- [14] D. Zeng, R. Veldhuis, and L. Spreeuwiers, "A survey of face recognition techniques under occlusion," arXiv, Jun. 2020.
- [15] N. Zhang et al., "Beyond Frontal Faces: Improving Person Recognition Using Multiple Cues," arXiv, Jan. 2015.
- [16] Y. Sun et al., "DeepID3: Face Recognition with Very Deep Neural Networks," arXiv, Feb. 2015.
- [17] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A Unified Embedding for Face Recognition and Clustering," Proc. IEEE CVPR, 2015.
- [18] P. Viola and M. Jones, "Rapid Object Detection using a Boosted Cascade of Simple Features," Proc. IEEE CVPR, 2001.
- [19] W. Deng et al., "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," Proc. IEEE/CVF CVPR, 2019.